



Nutrition & Digestion

CH 6 Science | Class 10

Notes + 10 Years Integrated PYQ's





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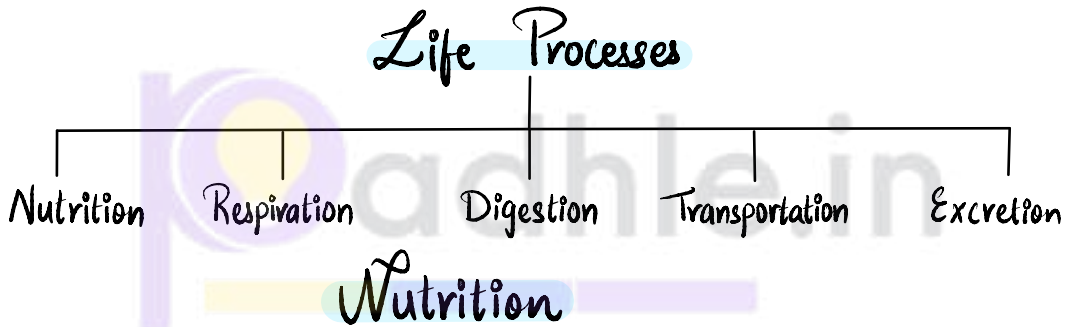


LIFE PROCESSES

What are life processes?

- The maintenance functions of living organisms must go on even when they are not doing anything particular.
- The processes which together perform this maintenance job are life processes.

What processes will we study about?



→ **Nutrients**: Group of food in which protein, fats, vitamins and minerals are involved is called nutrients.

→ **Nutrition**: The process of intake of nutrients (like carbohydrates, fats, proteins, minerals, vitamin and water) by an organism as well as the utilization of these nutrients by the organism.

Types of Nutrition

- i) Autotrophic nutrition
- ii) Heterotrophic (Holozoic) nutrition

i) Autotrophic Nutrition

→ It is a process where an organism prepares its own food.

- They are known as autotrophs.

Ex: green plants

ii) Heterotrophic nutrition

→ It is a type of nutrition in which organisms depend upon other organisms for food to survive.

Ex: animals, humans.

- In heterotrophic also, we see:

a) Saprophytic: Obtaining food from dead and decaying matter.

Ex: Fungi

b) Parasitic: Obtaining food from host organism without killing it.

Ex: Roundworm

c) Holozoic: Having solid matter and then digesting it is holozoic mode of nutrition.

Ex: Animals

Photosynthesis

♦ The process, by which green plants make their own food from carbon dioxide and water by using sunlight energy in the presence of chlorophyll, is called photosynthesis.

♦ Carbohydrates are utilised for providing energy to the plant.



• Oxygen is released during photosynthesis.

• The process of photosynthesis takes place in the green leaves of a plant.

• The food is prepared by the green leaves of a plant in the form of a simple sugar called glucose.

• The extra glucose is changed into another food called starch.

- This starch is stored in the leaves of the plant.
- The green plants convert sunlight energy into chemical energy by making carbohydrates.
- The carbon dioxide gas enters the leaves of the plant through the stomata present on their surface.

Obtaining water for photosynthesis

- The water required by the plants for photosynthesis is absorbed by the root of the plants from the soil through the process of osmosis.
- The water absorbed by the roots of the plants is transported upward through the xylem vessels to the leaves where it reaches the photosynthetic cells.

Site of photosynthesis

- The site of photosynthesis in a cell of the leaf are chloroplasts which contain chlorophyll.
- Chloroplasts are present in the photosynthetic cells (mesophyll cells) of green plants.
- These cells contain more chlorophyll than other plant cells.

Stomata

- 1) They allow the exchange of gasses (CO_2 and O_2) with the atmosphere.
- 2) Evaporation of water from the leaf surface occurs through the stomata.
 - Thus, the stomata help in the process of transpiration.
- 3) Allows the intake of carbon dioxide and to give out oxygen during the process of photosynthesis.

* How do organisms obtain their nutrition?

→ The process of nutrition involves the following:

* Ingestion → Digestion → Absorption → Assimilation → Egestion

Q U E S T I O N S

1. Why is diffusion insufficient to meet the oxygen requirements of multicellular organisms like humans?
2. What criteria do we use to decide whether something is alive?
3. What are outside raw materials used for by an organism?
4. What processes would you consider essential for maintaining life?

Answers

1) All the body cells are not in direct contact with the surrounding environment, like humans in multicellular organisms.

- Hence, each cell of the body does not get oxygen according to the requirement by the process of diffusion from environment.

- So, diffusion is insufficient to meet the oxygen demand of multicellular organisms.

2) Breathing and respiration are the main criteria used to decide whether something is alive.

- Though, living beings also show growth and movement.

3) An organism uses organic molecules as raw material.

- Whereas, food is used by Heterotrophs and carbon dioxide, minerals, water is used by autotrophs.

4) Following processes are essential for maintaining life:

- i) Nutrition
- ii) Respiration
- iii) Transportation
- iv) Excretion

NUTRITION IN AMOEBA

i) Ingestion

- Amoeba ingests food with the help of its finger-like extensions, called pseudopodia.
- When a food particle approaches Amoeba, it forms pseudopodia around it and forms a food vacuole inside the Amoeba.

ii) Digestion

- Various enzymes from the cytoplasm enter into the food vacuole and break them down into simple soluble molecules.

iii) Absorption

- The simple soluble food is absorbed by cytoplasm of Amoeba from food vacuoles through the process of diffusion.

iv) Assimilation

- Amoeba cell obtains energy from the absorbed food through respiration.
- This energy is utilised by Amoeba for its growth and repair of the body.

v) Egestion

- When a considerable amount of undigested food gets collected inside Amoeba, its cell membrane ruptures and throws out the undigested food.

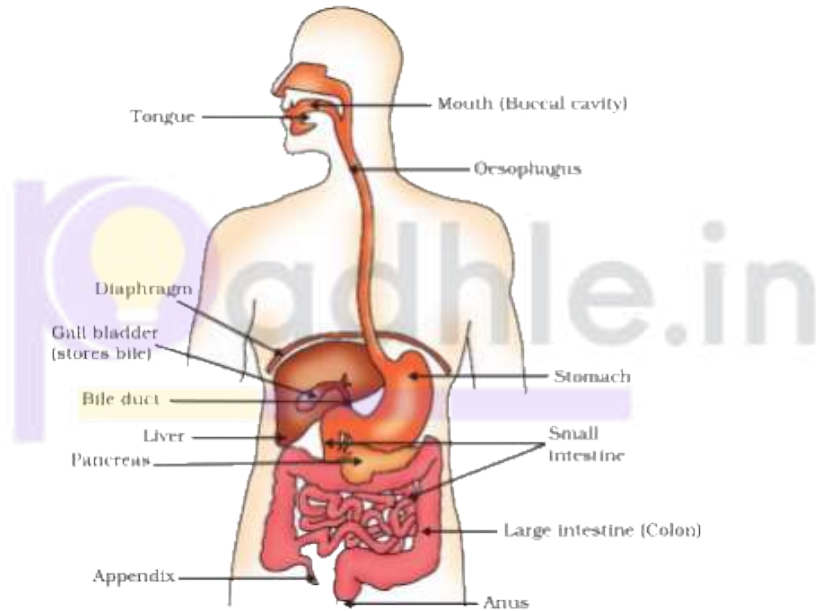
NUTRITION IN HUMAN BEINGS

* Important Definitions

① **Ingestion:** The process of intake of food by an organism is called Ingestion.

② **Digestion:** The process in which food containing large, insoluble molecules is broken down into small, water soluble molecules.

- ③ **Absorption**: The process in which the digested food passes through the intestinal wall into blood stream is called Absorption.
- ④ **Assimilation**: The process in which the absorbed food is taken in by body cells and used for energy, growth and repair is called Assimilation.
- ⑤ **Egestion (Excretion)**: The process in which the undigested food is removed from the body is called Egestion.



❖ Digestion begins in the mouth itself.

1) Mouth

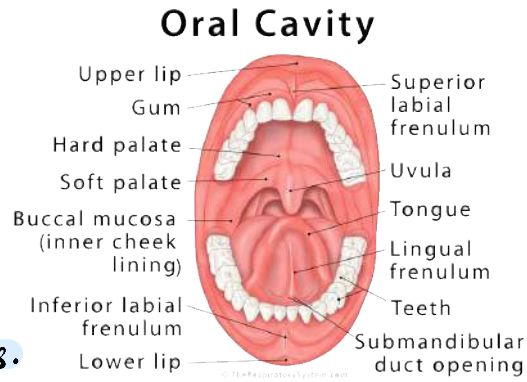
→ Digestion begins in the mouth itself.

- The saliva contains an enzyme called salivary amylase that breaks down starch which is a complex molecule into sugars like maltose.

2) Buccal or Oral Cavity

→ It extends from mouth to pharynx and lies between two jaws, upper (fixed)

- and lower (movable).
- It contains teeth also.
 - Teeth help in chewing and grinding the food.
 - Starch gets converted to simple sugar/maltose.
 - The movement of tongue turns food into mass of a substance called **BOLUS**.



3) Oesophagus

- No digestion of food takes place here.
- Peristaltic movement helps to pass the food down.

4) Stomach

- The muscular walls of the stomach help in mixing the food thoroughly with more digestive juices.
- These digestion functions are taken care by the gastric glands present in the wall of the stomach.

Gastric Glands

- Gastric Glands secrete gastric juice which consists of HCl, mucus and pepsinogen.

a) Function of HCl :

- 1) Converts inactive pepsinogen into active pepsin.
- 2) Kills the germs.

b) Function of Mucus:

- Protects the inner lining of stomach.

c) Function of Pepsin:

- Breaks down proteins into peptides.
- Pepsin is a major enzyme of gastric juice which functions in acidic medium.

- It hydrolyses proteins into soluble fragments called peptones and proteases.

- The exit of food from the stomach is regulated by a sphincter muscle which releases it in small amounts into the small intestine.

SMALL INTESTINE

- This is the longest part of the alimentary canal which is fitted into a compact space because of extensive coiling.
- Site of the complete digestion of carbohydrates, proteins and fats.
- The food coming from the stomach is acidic and has to be made alkaline for the pancreatic enzymes to act.
- * The two main functions are:

- 1) Emulsification of fats by Bile Juice which is secreted by liver.
 - 2) Pancreatic juice secreted by pancreas contains trypsin and lipase.
- Lipase: Breaks down emulsified fats.
- Trypsin: Helps to digest protein.

- The walls of the small intestine contain glands which secrete intestinal juice.
- The enzymes present in it finally convert the proteins to amino acids, complex carbohydrates into glucose and fats into fatty acids and glycerol.
- Following conversions take place in short:

Carbohydrates - Glucose Fats - Fatty acids + glycerol
Protein - Amino acids

- After that, the inner lining of the small intestine has numerous finger-like projections called villi which increase the surface area for absorption.
- The villi are richly supplied with blood vessels which take the absorbed food to each and every cell of the body, where it is utilised for obtaining energy, building up new tissues and the repair of old tissues.